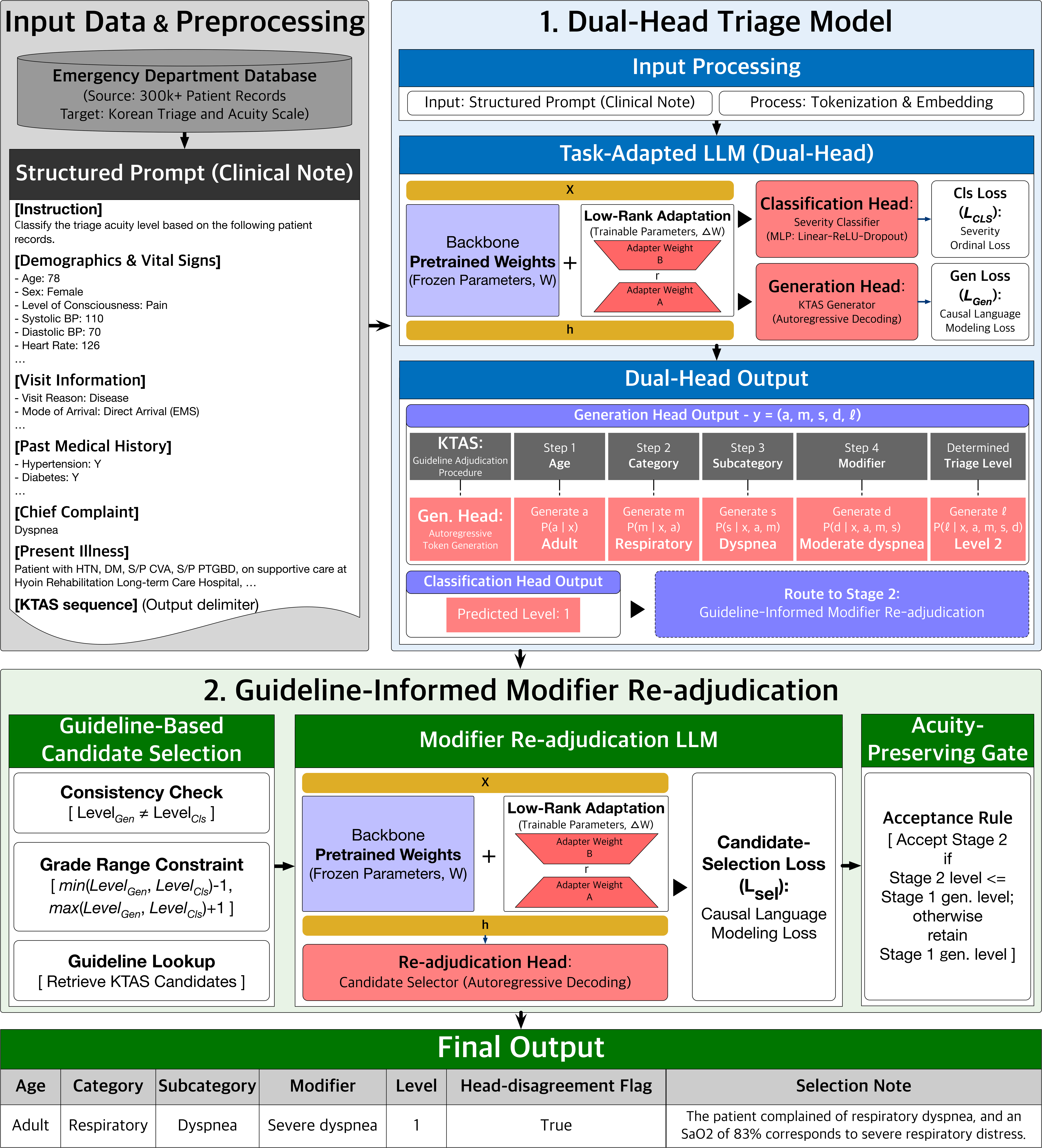
Input Data & Preprocessing
Emergency Department Database
(Source: 300k+ Patient Records
Target: Korean Triage and Acuity Scale)
Structured Prompt (Clinical Note)
[Instruction]
Classify the triage acuity level based on the following patient records.
[Demographics & Vital Signs]
- Age: 78
- Sex: Female
- Level of Consciousness: Pain
- Systolic BP: 110
- Diastolic BP: 70
- Heart Rate: 126
…
[Visit Information]
- Visit Reason: Disease
- Mode of Arrival: Direct Arrival (EMS)
…
[Past Medical History]
- Hypertension: Y
- Diabetes: Y
…
[Chief Complaint]
Dyspnea
[Present Illness]
Patient with HTN, DM, S/P CVA, S/P PTGBD, on supportive care at Hyoin Rehabilitation Long-term Care Hospital, …
[KTAS sequence] (Output delimiter)
1. Dual-Head Triage Model
Input Processing
Input: Structured Prompt (Clinical Note)
Process: Tokenization & Embedding
Task-Adapted LLM (Dual-Head)
x
Backbone Pretrained Weights (Frozen Parameters, W)
+
Low-Rank Adaptation (Trainable Parameters, ΔW)
Adapter Weight B
r
Adapter Weight A
h
Classification Head: Severity Classifier (MLP: Linear-ReLU-Dropout)
Cls Loss (L_{CLS}): Severity Ordinal Loss
Generation Head: KTAS Generator (Autoregressive Decoding)
Gen Loss (L_{Gen}): Causal Language Modeling Loss
Dual-Head Output
Generation Head Output - y = (a, m, s, d, ℓ)
KTAS: Guideline Adjudication Procedure
Step 1 Age
Step 2 Category
Step 3 Subcategory
Step 4 Modifier
Determined Triage Level
Gen. Head: Autoregressive Token Generation
Generate a P(a | x) Adult
Generate m P(m | x, a) Respiratory
Generate s P(s | x, a, m) Dyspnea
Generate d P(d | x, a, m, s) Moderate dyspnea
Generate ℓ P(ℓ | x, a, m, s, d) Level 2
Classification Head Output
Predicted Level: 1
Route to Stage 2: Guideline-Informed Modifier Re-adjudication
2. Guideline-Informed Modifier Re-adjudication
Guideline-Based Candidate Selection
Consistency Check [$Level_{Gen} \neq Level_{Cls}$]
Grade Range Constraint [$min(Level_{Gen}, Level_{Cls})-1$, $max(Level_{Gen}, Level_{Cls})+1$]
Guideline Lookup [Retrieve KTAS Candidates]
Modifier Re-adjudication LLM
x
Backbone Pretrained Weights (Frozen Parameters, W)
+
Low-Rank Adaptation (Trainable Parameters, ΔW)
Adapter Weight B
r
Adapter Weight A
h
Re-adjudication Head: Candidate Selector (Autoregressive Decoding)
Candidate-Selection Loss (L_{sel}): Causal Language Modeling Loss
Acuity-Preserving Gate
Acceptance Rule [Accept Stage 2 if Stage 2 level <= Stage 1 gen. level; otherwise retain Stage 1 gen. level]
Final Output
Age | Category | Subcategory | Modifier | Level | Head-disagreement Flag | Selection Note
Adult | Respiratory | Dyspnea | Severe dyspnea | 1 | True | The patient complained of respiratory dyspnea, and an SaO2 of 83% corresponds to severe respiratory distress.